Reliability

Time-Independent Reliabilities

- system composed of components such that when we examine the system, component i is working with probability of R_i (independent of time)
- components fail independently of other components

Components in Series

- the system is working if *all* of the components are working
- overall system reliability

$$R = R_1 \cdot R_2 \cdot R_3 \cdots R_n$$

Components in Parallel

- the system is working if *at least one* of the components is working
- in other words, system is failed if *all* components are failed:

$$1 - R = (1 - R_1)(1 - R_2) \cdots (1 - R_n)$$
$$R = 1 - (1 - R_1)(1 - R_2) \cdots (1 - R_n)$$

Triple Modular Redundancy

- system consists of three components
- each component has the same reliability, R
- components fail independently
- Here,

$$R_{TMR} = R^3 + \binom{3}{2} R^2 (1 - R)$$

Redundancy

- consider a number of components in series
- add a redundant component in parallel with one component
- for example, $R_1 = 0.95, R_2 = 0.9, R_3 = 0.95$
- $R = 0.81$

- in parallel with either component 1 or 3:

$$R = (0.95)(0.9)(1 - (1 - 0.95)^2) = 0.8529$$

- in parallel with component 2:

$$R = (0.95)^2(1 - (1 - 0.9)^2) = 0.8935$$

- in fact, improving component 2 is almost as good as improving both components 1 and 3 (where $R = 0.8955$)

Time-Dependent Reliabilities

- previous notion does not include the element of time
- how long does a system operate before failing?
- can a system become more/less likely to fail as time progresses

Lifetime

- X is a random variable that represents the lifetime of a component/system
- X has distribution $F(x)$ and density $f(x)$
- $f(t)\Delta t$ is approximately the probability that a failure occurs in the interval $(t, t + \Delta t]$
- would like to know the probability that a failure occurs in a (small) time interval, given that failure has not yet occurred
- make this precise using conditional probability

Failure Rate

- conditional likelihood of failure:

$$P\{X \leq t + x | X > t\} = \frac{P\{t < X \leq t + x\}}{P\{X > t\}} = \frac{F(t + x) - F(t)}{1 - F(t)}$$

- from this, we calculate the instantaneous failure rate, $h(t)$:

$$h(t) = \lim_{x \to 0} \frac{1}{x} \frac{F(t + x) - F(t)}{1 - F(t)}$$
$$= \frac{f(t)}{1 - F(t)}$$

Failure Rate

- Interpretation: $h(t)\Delta t$ is conditional probability that a component that has survived to age t will fail in the interval $(t, t + \Delta t]$
- $h(t)$ is also known as the *hazard rate*

Some Relations

- Suppose that you are given $h(t)$ rather than $F(t)$
- With the notation $R(t) = P\{X > t\}$, we can derive

$$R(t) = exp\left[-\int_0^t h(x)dx\right]$$
$$F(t) = 1 - R(t)$$

Rated Lifetime

- The age that 90 percent of components are expected to exceed
- In other words, if L is the rated lifetime, then $F(L) = 0.1$

Example

The following is data for a liquid cooling cartridge system used in Sun enterprise-class servers (10-15 years ago). There are three components, a blower, a water pump, and a compressor. The lifetimes of each are independent and follow Weibull distributions (these are very popular for reliability modelling). A Weibull distribution is described by two parameters, λ and α, with the density given by

$$f(t) = \lambda \alpha t^{\alpha-1} e^{-\lambda t^{\alpha}}, \qquad t \geq 0$$

From the density, we can derive

$$F(t) = 1 - e^{-\lambda t^{\alpha}}$$
$$h(t) = \lambda \alpha t^{\alpha - 1}$$

The data for each of the components is α and the *rated life*. For each component, $\alpha = 3.0$, while the rated life for the blower is 70,000 hours, for the other two 100,000 hours.

IFR and DFR

- IFR (Increasing Failure Rate) - $h(t)$ is a nondecreasing function of t
- IFR – more likely to fail as they age
- DFR (Decreasing Failure Rate) - $h(t)$ is nonincreasing function of t
- DFR – less likely to fail as they age
- In general, distributions may be neither IFR nor DFR
- so-called “bathtub curve”
- range over which failure rate is constant is quite often the useful life

IFR and DFR - Implications

- DFR – never preemptively replace a component
- IFR – may become desirable to preemptively replace a component before it fails

Constant Failure Rate

- corresponds to an exponential distribution
- suppose lifetime is exponentially distributed with rate λ

$$
\begin{aligned}
h(t) &= \frac{f(t)}{1 - F(t)} \\
&= \frac{\lambda e^{-\lambda t}}{e^{-\lambda t}} \\
&= \lambda
\end{aligned}
$$

CTMC approach

- with constant failure rates, CTMC approach is applicable
- already looked at some reliability models
- useful for answering questions like the availability of resources (what is the probability that there are more than n servers working)

Machine Repair Problem

- N machines, each fail independently with rate λ
- M servers for repairs, each work on a single machine with repair rate μ
- This is a birth-death process, proceed as with CTMC material

Machine Repair Problem - Example

Consider a machine repair problem with three machines. Assume $\lambda = .001$ and $\mu = 1$. Calculate the probability that all three machines are working with both one server and two servers.

Machine Repair Problem - Assumptions

- repair times exponentially distributed could be a problematic assumption
- turns out to be not too sensitive as long as repair times are (much) less than failure times

Mean Time to Failure

- may be interested in mean time to failure, for example in the previous problem what is the mean time from all machines working to no machines working (or any other state)?
- usual CTMC approach not appropriate – if we consider a terminal state, steady-state analysis fails
- however, can solve the problem if we define $T_{i,j}$ to be the time to go from state i to state j, $\frac{1}{\gamma_i}$ to be the expected time spent in state i, and P_{ij} to be the probability that the next state after state i is state j

- define state 0 to be the initial state, state N to be the final state, define MTTF as the time to go between state 0 and N
- For each state $j = 0, \dots, N-1$, write

$$E\left[T_{j,N}\right] = \frac{1}{\gamma_j} + \sum_{k=0}^{N-1} P_{jk} E\left[T_{k,N}\right]$$

- gives a system of N equations in N unknowns – we are interested in $E\left[T_{0,N}\right]$

Mean Time to Failure Example

- This type of model is used in a variety of settings – a system has three states (operational, suboptimal, outage)
- Critical failures leading to outage occur at rate λ_c when either operational or suboptimal
- Normal degradation (operational to suboptimal, suboptimal to outage) at rate λ_d
- Repair from suboptimal at rate μ_d
- Repair from outage to optimal at rate μ_c
- Interested in MTTF